

ALPCAHUS: Subspace Clustering for Heteroscedastic Data

Javier Salazar Cavazos[✉], *Graduate Student Member, IEEE*,

Jeffrey A. Fessler[✉], *Fellow, IEEE*, and Laura Balzano[✉], *Senior Member, IEEE*

EECS Department, University of Michigan, Ann Arbor, Michigan, United States

Email: {javiersc, fessler, girasole}@umich.edu

Abstract—Principal component analysis (PCA) is a key tool in the field of data dimensionality reduction. Various methods have been proposed to extend PCA to the union of subspace (UoS) setting for clustering data that come from multiple subspaces like K -Subspaces (KSS). However, some applications involve heterogeneous data that vary in quality due to noise characteristics associated with each data sample. Heteroscedastic methods aim to deal with such mixed data quality. This paper develops a heteroscedastic-focused subspace clustering method, named ALPCAHUS, that can estimate the sample-wise noise variances and use this information to improve the estimate of the subspace bases associated with the low-rank structure of the data. This clustering algorithm builds on K -Subspaces (KSS) principles by extending the recently proposed heteroscedastic PCA method, named LR-ALPCAH, for clusters with heteroscedastic noise in the UoS setting. Simulations and real-data experiments show the effectiveness of accounting for data heteroscedasticity compared to existing clustering algorithms. Code available at <https://github.com/javiersc1/ALPCAHUS>.

Index Terms—Heteroscedastic data, heterogeneous data quality, subspace bases estimation, subspace clustering, union of subspace model, unsupervised learning.

I. INTRODUCTION

Many modern data science problems require learning an approximate signal subspace basis for some collection of data. This is important for downstream tasks involving subspace basis coefficients such as classification [1], regression [2], and compression [3]. Besides subspace learning, one may be interested in clustering data points that originate from multiple subspaces. Formally, subspace clustering, or union of subspace (UoS) modeling, is an unsupervised machine learning problem where the goal is to cluster unlabeled data and find the subspaces associated with each data cluster. When the cluster assignments are known, it is easy to find the subspaces, and vice versa. This problem becomes nontrivial when both components must be estimated [4]. This clustering problem has many applications, such as image segmentation [5], motion segmentation [6], image compression [7], and system identification [8].

Some applications involve heterogeneous data samples that vary in quality due in part to noise characteristics associated with each sample. Some examples of heteroscedastic datasets include environmental air quality data [9], astronomical spectral data [10], and biological sequencing data [11]. In heteroscedastic settings, the noisier data samples can significantly corrupt the basis estimates [12]. In turn, this corruption can worsen

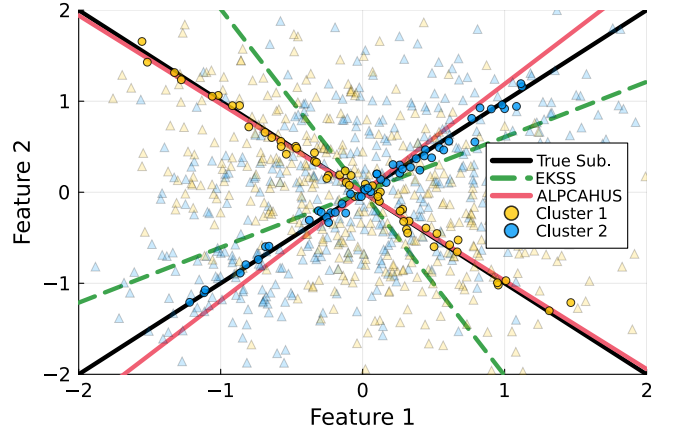


Fig. 1: Two 1D subspaces, colored blue and yellow, with data consisting of two noise groups shown with circle and triangle marker types.

clustering performance. Popular clustering methods such as Sparse Subspace Clustering (SSC) [13], K -Subspaces (KSS) [14], and Subspace Clustering via Thresholding (TSC) [15] implicitly assume that data quality is consistent. For example, in SSC, the method relies on the self-expressiveness property of data that requires using other samples to estimate similarity. From our experiments, we found that this implicit data quality assumption can degrade clustering quality for heteroscedastic data.

Because of these limitations, we developed a subspace clustering algorithm, inspired from KSS principles, that explicitly models noise variance terms, without assuming that data quality is known. The method adaptively clusters data while learning noise characteristics. See Fig. 1 for a visualization where Ensemble KSS (EKSS) [16] returns poor subspace bases estimates whereas our method found more accurate subspace bases and improved clustering quality.

We extend our previous work [17] by generalizing the LR-ALPCAH formulation to the UoS setting for clustering heteroscedastic data. The proposed approach achieved ~ 3 times lower clustering error than existing methods, and it achieved relatively low clustering error even when very few high quality samples are available. The paper is divided into a few key sections. Section II introduces the heteroscedastic problem formulation for subspace clustering. Section III discusses related work in subspace clustering and reviews the heteroscedastic

subspace algorithm LR-ALPCA. Section IV introduces the proposed subspace clustering method named ALPCAUS. Section V covers synthetic and real data experiments that illustrate the effectiveness of modeling heteroscedasticity in a clustering context. Sec. VI discusses some limitations of our method and possible extensions.

II. PROBLEM FORMULATION

Let K denote the number of subspaces that is either known beforehand or estimated using other methods. Before describing the general union of subspace model, we consider the single subspace model when $K = 1$.

A. Single-Subspace Model ($K = 1$)

Let $\mathbf{y}_i \in \mathbb{R}^D$ denote the data samples for index $i \in \{1, \dots, N\}$, where D denotes the ambient dimension. Let $\mathbf{x}_i$ represent the low-dimensional data sample generated by $\mathbf{x}_i = \mathbf{U}\mathbf{z}_i$ where $\mathbf{U} \in \mathbb{R}^{D \times d}$ is an unknown basis for a subspace of dimension d and $\mathbf{z}_i \in \mathbb{R}^d$ are the corresponding basis coordinates. Then the heteroscedastic model we consider is

$$\mathbf{y}_i = \mathbf{x}_i + \boldsymbol{\epsilon}_i \quad \text{where} \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \nu_i \mathbf{I}) \quad (1)$$

assuming Gaussian noise with variance ν_i , where $\mathbf{I}$ denotes the $D \times D$ identity matrix. We consider two cases: one where each data sample may have its own noise variance, and one where the case where there are G groups of data having shared noise variance terms $\{\nu_1, \dots, \nu_G\}$. Sec. III-B discusses an optimization problem based on this model that estimates the heterogeneous noise variances $\{\nu_i\}$ and the subspace basis $\mathbf{U}$.

B. Union of Subspaces Model ($K \geq 1$)

Let $\mathbf{Y} = [\mathbf{y}_1 \dots \mathbf{y}_N] \in \mathbb{R}^{D \times N}$ denote a matrix whose columns consist of all N data points $\mathbf{y}_i \in \mathbb{R}^D$. We generalize (1) to model the data with a union of subspaces model as follows

$$\begin{aligned} \mathbf{y}_i &= \mathbf{x}_i + \boldsymbol{\epsilon}_i \\ \mathbf{x}_i &= \mathbf{U}_{k_i} \mathbf{z}_i \text{ for some } k_i \in \{1, \dots, K\}, \end{aligned} \quad (2)$$

where $\mathbf{U}_k \in \mathbb{R}^{D \times d_k}$ is a subspace basis that has subspace dimension d_k . Here $\mathbf{z}_i \in \mathbb{R}^{d_k}$ denotes the basis coefficients associated with $\mathbf{x}_i$, and $\boldsymbol{\epsilon}_i \in \mathbb{R}^D$ denotes noise for that point drawn from $\boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \nu_i \mathbf{I})$.

If the subspace bases were known, then one would like to find the associated subspace label $c_i \in \{1, \dots, K\}$ for each data sample by solving the following optimization problem

$$c_i = \arg \min_k \|\mathbf{y}_i - \mathbf{U}_k \mathbf{U}_k^T \mathbf{y}_i\|_2^2, \quad \forall \mathbf{y}_i \in \mathbf{Y}. \quad (3)$$

This label describes the subspace association of a data point $\mathbf{y}_i$ that has the lowest residual between the original sample and its reconstructed value given the subspace model. In general, the goal is to estimate all of the subspace bases $\mathcal{U} = (\mathbf{U}_1, \dots, \mathbf{U}_K)$ and cluster assignments $\mathcal{C} = (c_1, \dots, c_N)$ that describe the subspace labels of all points.

III. RELATED WORKS

A. Subspace Clustering

Many subspace clustering algorithms fall into a general umbrella of categories such as algebraic methods [18], iterative methods [19], statistical methods [20], and spectral clustering methods [21] [22]. In recent years, both spectral clustering-based methods and iterative methods have become popular.

1) *Spectral Clustering*: Many methods build on spectral clustering. This method is often used for clustering nodes in graphs. Spectral clustering aims to find the minimum cost ‘‘cuts’’ in the graph to partition nodes into clusters. Given some collection of data points, one way to construct a graph is to assume that ‘‘nearby’’ data samples are highly connected in the graph. Thus, it is possible to construct a graph from data samples, with various levels of connectedness, by using some metric to assign affinity/similarity for each pair of points. Let $\mathcal{G}(\mathcal{V}, \mathbf{W})$ correspond to a graph that consists of vertices $\mathcal{V} = \{v_1, \dots, v_N\}$ and edge weights $\mathbf{W} \in \mathbb{R}^{N \times N}$ such that w_{ij} corresponds to some nonnegative weight, or similarity, between v_i and v_j . The graph $\mathcal{G}$ is assumed to be undirected, i.e., $\mathbf{W} = \mathbf{W}^T$. The degree of a node is defined as $d_i = \sum_{j=1}^N w_{ij}$ and can be collected to form a degree matrix such that $\mathbf{D} = \text{diag}(d_1, \dots, d_N)$. Let $\mathcal{A}_1, \dots, \mathcal{A}_K$ form a K -partition on $\mathcal{G}(\mathcal{V}, \mathbf{W})$ and $\text{cut}(\mathcal{A}, \mathcal{B}) = \sum_{i \in \mathcal{A}, j \in \mathcal{B}} w_{ij}$. Then, spectral clustering aims to solve the following optimization problem:

$$\arg \min_{\mathcal{A}_1, \dots, \mathcal{A}_K} \frac{1}{2} \sum_{i=1}^K \frac{\text{cut}(\mathcal{A}_i, \bar{\mathcal{A}}_i)}{|\mathcal{A}_i|} \quad (4)$$

where $\bar{\mathcal{A}}_i$ is all vertices not belonging in $\mathcal{A}_i$ and $|\mathcal{A}_i|$ is the number of vertices belonging to the i th partition. Instead of working with $\mathbf{W}$ directly, one computes a normalized graph Laplacian such as $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1} \mathbf{W}$ to make the influence of heavy degree nodes more similar to low degree nodes. Collecting the indicator values

$$h_{ij} \triangleq \begin{cases} 1/\sqrt{|\mathcal{A}_j|} & \text{if } v_i \in \mathcal{A}_j \\ 0 & \text{otherwise} \end{cases}$$

into a matrix $\mathbf{H} \in \mathbb{R}^{N \times K}$, one can rewrite (4) as

$$\frac{1}{2} \sum_{i=1}^K \frac{\text{cut}(\mathcal{A}_i, \bar{\mathcal{A}}_i)}{|\mathcal{A}_i|} = \sum_{i=1}^K (\mathbf{H}^T \mathbf{L} \mathbf{H})_{ii} = \text{Tr}(\mathbf{H}^T \mathbf{L} \mathbf{H}), \quad (5)$$

where $\text{Tr}(\mathbf{H}) = \sum_i \mathbf{H}_{ii}$ denotes the trace of a matrix. The indicator vectors are discrete, which makes the problem computationally challenging. One way to relax the problem is to allow arbitrary real values for $\mathbf{H}$ and instead solve

$$\hat{\mathbf{H}} = \arg \min_{\mathbf{H} \in \mathbb{R}^{N \times K}} \text{Tr}(\mathbf{H}^T \mathbf{L} \mathbf{H}) \text{ s.t. } \mathbf{H}^T \mathbf{H} = \mathbf{I}. \quad (6)$$

Trace minimization problems with semi-unitary constraints are well-studied in the literature, and it is easy to see that $\hat{\mathbf{H}}$ consists of the first K eigenvectors of $\mathbf{L}$ assuming that $\lambda_1(\mathbf{L}) \leq \dots \leq \lambda_N(\mathbf{L})$ where $\lambda_i(\mathbf{L})$ denotes the i th eigenvalue of $\mathbf{L}$. Once $\hat{\mathbf{H}}$ is computed, the K -means algorithm is applied to $\hat{\mathbf{H}}^T$, treating each row of $\hat{\mathbf{H}}$ as the spectral embedding of the associated v_i vertex. We now describe three subspace clustering methods that use this technique by first forming a

weight matrix $\mathbf{W}$ and then applying the spectral clustering method.

2) *Self-Expressive Methods*: Self-expressive methods exploit the “self-expressiveness property” [23] of data that hypothesizes that a single data point can be expressed as a linear combination of other data points in its cluster, which is trivially true if the data exactly follows a subspace model. The goal is to learn those linear coefficients and which is often achieved by adopting different regularizers in the formulation. These self-expressive algorithms, e.g., SSC [13] and LRSC [24], learn the coefficient matrix $\mathbf{P} \in \mathbb{R}^{N \times N}$ by solving special cases of the following general optimization problem

$$\arg \min_{\mathbf{P}} f(\mathbf{Y} - \mathbf{Y}\mathbf{P}) + \lambda \mathcal{R}(\mathbf{P}) \quad \text{s.t. } \mathbf{P} \in \mathcal{S}_{\mathbf{P}}. \quad (7)$$

The function $f(\mathbf{Y} - \mathbf{Y}\mathbf{P})$ is a data fidelity term, $\mathcal{R}(\mathbf{P})$ is a regularizer, and $\mathcal{S}_{\mathbf{P}}$ is some constrained set to encourage $\mathbf{P}$ to satisfy certain conditions. In the case of SSC the optimization problem is

$$\arg \min_{\mathbf{P}} \|\mathbf{Y} - \mathbf{Y}\mathbf{P}\|_{\text{F}}^2 + \lambda \|\mathbf{P}\|_{1,1} \quad \text{s.t. } \mathbf{P}_{ii} = 0 \quad \forall i \quad (8)$$

where $\|\cdot\|_{1,1} = \|\text{vec}(\cdot)\|_1$ is the vectorized 1-norm and $\|\cdot\|_{\text{F}}$ is the Frobenius norm of a matrix. Observe that (8) approximates each data sample as a sparse linear combination of other data points to form $\mathbf{P}$. Let $\text{abs}(\mathbf{P})$ represent the element-wise absolute value of matrix $\mathbf{P}$. Then, spectral clustering is performed on $\mathbf{W} = \frac{1}{2}(\text{abs}(\mathbf{P}^T) + \text{abs}(\mathbf{P}))$ by applying the K -means method to the spectral embedding of the affinity matrix $\mathbf{W}$. Ideally, (8) would select the high quality data to represent worse samples in $\mathbf{P}$ and this information would be retained in $\mathbf{W}$. However, this data quality awareness condition is not guaranteed in self-expressive methods to our knowledge. In our experiments, Fig. 2f and Fig. 6 show that there must be an issue constructing an ideal $\mathbf{P}$ due to poor clustering performance in heteroscedastic conditions.

3) *Subspace Clustering via Thresholding (TSC)*: In the example to follow, assume that there exists two unique 1-dimensional subspaces. Given two data samples $\mathbf{y}_i$ and $\mathbf{y}_j$, intuitively their dot product $\langle \mathbf{y}_i, \mathbf{y}_j \rangle$ will be high if $c_i = c_j$ meaning they belong to the same subspace. Likewise, if $c_i \neq c_j$, then $\langle \mathbf{y}_i, \mathbf{y}_j \rangle = 0$ assuming orthogonal subspaces. In non-orthogonal scenarios, $\langle \mathbf{y}_i, \mathbf{y}_j \rangle$ will be relatively small since $c_i \neq c_j$ as compared to $c_i = c_j$. Using this idea, in more general D -dimensional subspace settings, TSC constructs a matrix $\mathbf{Z} \in \mathbb{R}^{N \times N}$ such that

$$\mathbf{Z}_{ij} = \exp(-2 \arccos(|\langle \mathbf{y}_i, \mathbf{y}_j \rangle|)) \quad \text{s.t. } \mathbf{Z}_{ii} = 0 \quad \forall i. \quad (9)$$

This matrix is then thresholded to retain only the top q values for each row, i.e.,

$$\mathbf{W} = \arg \min_{\mathbf{W}} \|\mathbf{W} - \mathbf{Z}\|_{\text{F}}^2 \quad \text{s.t. } \|\mathbf{W}_{i,:}\|_0 = q \quad \forall i \quad (10)$$

where $\|\mathbf{W}_{i,:}\|_0 = q$ is the l_0 pseudo-norm. In other words, one constructs $\mathbf{W}_{i,:}$ using the q nearest neighbors that correspond to the highest magnitude values. Then, spectral clustering is applied to $\mathbf{W}$ to find the subspace clustering associations.

4) *Ensemble K -Subspaces (EKSS)*: Iterative methods are based on the idea of alternating between cluster assignments and subspace basis approximation, with KSS being highly prominent [25]. The KSS algorithm seeks to solve the following optimization problem:

$$\arg \min_{\mathcal{C}, \mathcal{U}} \sum_{k=1}^K \sum_{c_i=k, \forall i} \|\mathbf{y}_i - \mathbf{U}_k \mathbf{U}_k^T \mathbf{y}_i\|_2^2. \quad (11)$$

The goal is to minimize the sum of residual norms by alternating between performing PCA on each cluster to update $\mathcal{U}$ and using the subspace bases to calculate new cluster assignments $\mathcal{C}$ in a similar fashion to the K -means algorithm.

The quality of the solution depends highly on the initialization. Recent work provides convergence guarantees and spectral initialization schemes that provably perform better than random initialization [26]. However, this problem (11), as shown in [27], is NP-hard. Further, it is prone to local minima [28]. To overcome this, consensus clustering [29] is a tool that leverages information from many trials and combines results together. This approach, known as Ensemble KSS (EKSS) [16] in this context, creates an affinity matrix whose (i, j) th entry represents the number of times the two points were clustered together in a trial. Then, spectral clustering is performed on the affinity matrix to get the final clustering from the many base clusterings. The use of PCA makes it challenging to learn clusters and subspace bases in the heteroscedastic regime since PCA implicitly assumes the same noise variance across all samples [30]. The proposed ALPCAUS approach in Sec. IV builds on KSS, and its ensemble version, by generalizing (11) to the heteroscedastic regime.

B. Single Subspace Heteroscedastic PCA Method

Before extending (11) to the heteroscedastic setting, we review how LR-ALPCAUS [17] solves for a single subspace ($K = 1$) in the heteroscedastic setting using the model described in (1). For the measurement model $\mathbf{y}_i \sim \mathcal{N}(\mathbf{x}_i, \nu_i \mathbf{I})$ in (1), the probability density function for a single data sample $\mathbf{y}_i$ is

$$\frac{1}{\sqrt{(2\pi)^D |\nu_i \mathbf{I}|}} \exp \left[-\frac{1}{2} (\mathbf{y}_i - \mathbf{x}_i)' (\nu_i \mathbf{I})^{-1} (\mathbf{y}_i - \mathbf{x}_i) \right]. \quad (12)$$

After further manipulation, the negative log-likelihood is expressed in matrix notation as

$$\frac{1}{2} \|(\mathbf{Y} - \mathbf{X}) \mathbf{\Pi}^{-1/2}\|_{\text{F}}^2 + \frac{D}{2} \log |\mathbf{\Pi}|. \quad (13)$$

To reduce computation and enforce a low-rank solution, LR-ALPCAUS [17] took inspiration from the matrix factorization literature [31] and factorized $\mathbf{X} \in \mathbb{R}^{D \times N} \approx \mathbf{L} \mathbf{R}'$ where $\mathbf{L} \in \mathbb{R}^{D \times \hat{d}}$ and $\mathbf{R}' \in \mathbb{R}^{N \times \hat{d}}$ for some rank estimate $\hat{d}$. Using this idea, LR-ALPCAUS estimates $\mathbf{X}$ by solving for $\mathbf{L}$ and $\mathbf{R}$ in the following optimization problem

$$\arg \min_{\mathbf{L}, \mathbf{R}, \mathbf{\Pi}} \frac{1}{2} \|(\mathbf{Y} - \mathbf{L} \mathbf{R}') \mathbf{\Pi}^{-1/2}\|_{\text{F}}^2 + \frac{D}{2} \log |\mathbf{\Pi}|. \quad (14)$$

The next section, Sec. IV, combines ideas from (11) and (14) to tackle the general union of subspaces setting ($K > 1$).

C. Other heteroscedastic models

This paper focuses on heteroscedastic noise across the data samples. There are other subspace learning methods in the literature that explore heteroscedasticity in different ways. For example, HeteroPCA considers heteroscedasticity across the feature space [32]. One possible application of that model is for data that consists of sensor information with multiple devices that naturally have different levels of precision and signal to noise ratio (SNR). Another heterogeneity model considers the noise to be homoscedastic and instead assumes that the signal itself is heteroscedastic [33]. That work considers clustering applications where the power fluctuating (i.e., heteroscedastic) signals are embedded in white Gaussian noise. We exclude comparisons with those methods since their models are very different. These different models each have their own applications.

IV. PROPOSED SUBSPACE CLUSTERING METHOD

For notational simplicity, let $\mathbf{Y}_k \triangleq \mathbf{Y}_{C_k} \in \mathbb{R}^{D \times N_k}$ denote the submatrix of $\mathbf{Y}$ having columns corresponding to data samples that are estimated to belong in the k th subspaces, i.e., $\mathbf{Y}_{C_k} = \text{matrix}(\{\mathbf{y}_i : c_i = k\})$. We apply this notation similarly to other matrices such as $\mathbf{\Pi}_k \triangleq \mathbf{\Pi}_{C_k} = \text{diag}(\{\nu_i : c_i = k\})$. For the union of subspace measurement model (2), we generalize LR-ALPCAH (14) with the following optimization problem

$$\arg \min_{\mathcal{L}, \mathcal{R}, \mathbf{\Pi}, \mathcal{C}} \sum_{k=1}^K \frac{1}{2} \left\| (\mathbf{Y}_k - \mathbf{L}_k \mathbf{R}_k^T) \mathbf{\Pi}_k^{-1/2} \right\|_F^2 + \frac{D}{2} \log |\mathbf{\Pi}_k| \quad (15)$$

where $\mathcal{C}, \mathbf{\Pi}, \mathcal{L}, \mathcal{R}$ denote the sets of estimated clusters, noise variances, and factorized matrices respectively for each cluster $k = 1, \dots, K$. Specifically $\mathcal{L} \triangleq \{\mathbf{L}_1, \dots, \mathbf{L}_K\}$, $\mathcal{R} \triangleq \{\mathbf{R}_1, \dots, \mathbf{R}_K\}$. Our algorithm for solving (15) is called ALPCAHUS (ALPCAH for Union of Subspaces). Similar to KSS, we alternate between updating subspace bases given cluster estimates and updating cluster estimates by projection given subspace bases estimates. We solve (15) using alternating minimization between

$$\arg \min_{\mathbf{L}_k, \mathbf{R}_k, \mathbf{\Pi}_k} \frac{1}{2} \left\| (\mathbf{Y}_k - \mathbf{L}_k \mathbf{R}_k^T) \mathbf{\Pi}_k^{-1/2} \right\|_F^2 + \frac{D}{2} \log |\mathbf{\Pi}_k|, \forall k \quad (16)$$

and

$$c_i = \arg \min_k \left\| \mathbf{y}_i - \mathbf{U}_k \mathbf{U}_k^T \mathbf{y}_i \right\|_2^2, \quad i = 1, \dots, N. \quad (17)$$

The solution to (16) is described in [17], with convergence theory. For completeness, we include the updates here. For the

$(i+1)$ iteration, given current (i) th iterates, the updates are given as

$$\begin{aligned} \mathbf{L}_k^{(i+1)} &= \arg \min_{\mathbf{L}_k} f(\mathbf{L}_k, \mathbf{R}_k^{(i)}, \mathbf{\Pi}_k^{(i)}) \\ &= \mathbf{Y}_k \mathbf{\Pi}_k^{(i)} \mathbf{R}_k^{(i)} (\mathbf{R}_k^{(i)T} \mathbf{\Pi}_k^{(i)} \mathbf{R}_k^{(i)})^{-1} \end{aligned} \quad (18)$$

$$\begin{aligned} \mathbf{R}_k^{(i+1)} &= \arg \min_{\mathbf{R}_k} f(\mathbf{L}_k^{(i)}, \mathbf{R}_k, \mathbf{\Pi}_k^{(i)}) \\ &= \mathbf{Y}_k^T \mathbf{L}_k^{(i)} (\mathbf{L}_k^{(i)T} \mathbf{L}_k^{(i)})^{-1} \end{aligned} \quad (19)$$

$$\begin{aligned} \mathbf{\Pi}_k^{(i+1)} &= \arg \min_{\mathbf{\Pi}_k} f(\mathbf{L}_k^{(i)}, \mathbf{R}_k^{(i)}, \mathbf{\Pi}_k) \implies \\ e_j^T \mathbf{\Pi}_k^{(i+1)} e_j &= |\mathcal{C}_k|^{-1} \|(\mathbf{Y}_k - \mathbf{L}_k^{(i)} \mathbf{R}_k^{(i)T}) e_j\|_2^2, \quad \forall j, \end{aligned} \quad (20)$$

where e_j denotes the j th standard canonical basis vector that is used to select the j th column of some matrix. To further improve clustering performance, we leverage consensus clustering over many trials. Initially, we tried using this approach with only one trial as seen in Fig. 2b and ALPCAHUS ($B = 1$) result in Fig. 6a. However, since K -subspaces in general is sensitive to initialization, we found great success in using a consensus approach with more than one trials as shown in Fig. 2d and ALPCAHUS ($B = 16$) result in Fig. 6a.

A. Ensemble Extension for ALPCAHUS

We now present the ensemble algorithm with base clustering parameter B to combine multiple trials of finding $\mathcal{C}$ in (15). Any $B > 1$ leverages consensus clustering by forming an affinity matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ where

$$\mathbf{W}_{i,j} = \frac{1}{B} \left| \{ \forall b \in \{1, \dots, B\} : \mathbf{y}_i, \mathbf{y}_j \text{ co-clustered in } \mathcal{C}^{(b)} \} \right| \quad (21)$$

and $\mathcal{C}^{(b)} = \{c_1^{(b)}, \dots, c_N^{(b)}\}$ refers to the cluster labels for the b th trial ranging from 1 to B . Then, the rows and columns of $\mathbf{W}$ are thresholded to retain the top q values by solving

$$\mathbf{Z}^{\text{row}} = \arg \min_{\mathbf{Z}} \|\mathbf{W} - \mathbf{Z}\|_F^2 \quad \text{s.t.} \quad \|\mathbf{W}_{i,:}\|_0 = q \quad \forall i \quad (22)$$

$$\mathbf{Z}^{\text{col}} = \arg \min_{\mathbf{Z}} \|\mathbf{W} - \mathbf{Z}\|_F^2 \quad \text{s.t.} \quad \|\mathbf{W}_{:,i}\|_0 = q \quad \forall i. \quad (23)$$

Finally, spectral clustering is applied to $\frac{1}{2}(\mathbf{Z}^{\text{col}} + \mathbf{Z}^{\text{row}})$ to get the clusters from the results of the ensemble. One could select the base clusterings parameter $B = 1$ to reduce to one trial of the optimization problem (15). Alg. ALPCAHUS summarizes the procedure for subspace clustering with optional ensemble learning; for Julia code implementations see <https://github.com/javiersci/ALPCAHUS>.

B. Rank Estimation

In subspace clustering, some algorithms like SSC do not require subspace dimension to be known or estimated, whereas others like KSS require this. For the group of algorithms that require this parameter to be estimated, there is great interest in adaptive methods that can learn the subspace dimension. In recent work, [26] proposes using an eigengap heuristic on the sample covariance matrix of each cluster, i.e., $S_k = \frac{1}{|\mathcal{C}_k|} \mathbf{Y}_k \mathbf{Y}_k^T$, to estimate dimension. This means calculating the following

$$\hat{d}_k = \arg \max_i |\lambda_i(S_k) - \lambda_{i+1}(S_k)| \quad (24)$$

Algorithm ALPCAUS (unknown noise variances, unknown noise grouping)

Input: $\mathbf{Y} \in \mathbb{R}^{D \times N}$: data, $K \in \mathbb{Z}^+$: number of subspaces, $\{\hat{d}_k \in \mathbb{Z}^+ \ \forall k \in \{1, \dots, K\}\}$: candidate dimension for all clusters, $q \in \mathbb{Z}^+$: threshold parameter, $B \in \mathbb{Z}^+$: number of base clusterings, $I \in \mathbb{Z}^+$: number of ALPCAUS iterations, $T \in \mathbb{Z}^+$: number of alternating updates

Output: $\mathcal{C} = \{c_1, \dots, c_K\}$: clusters of $\mathcal{Y}$

for $b = 1, \dots, B$ **(in parallel)** **do**
 $\mathcal{C}_k \sim \{1, \dots, N\}$ s.t. $|\mathcal{C}_k| \approx \frac{N}{K}$ for $k = 1, \dots, K$ Initialize clusters randomly without replacement
 for $k = 1, \dots, K$ **(in parallel)** **do**
 $\mathbf{L}_k, \mathbf{R}_k, \boldsymbol{\nu}_k \leftarrow \text{LR-ALPCAUS}(\mathbf{Y}_k, \hat{d}_k, I)$
 $\boldsymbol{\Pi}_k^{-1} \leftarrow \text{Diagonal}(\{\nu^{-1} : \nu \in \boldsymbol{\nu}_k\})$ Initialize $\mathbf{L}, \mathbf{R}$ matrices and form noise variance matrix
 end for
 for $t = 1, \dots, T$ **(in sequence)** **do**
 for $k = 1, \dots, K$ **(in parallel)** **do**
 $\mathbf{L}_k, \mathbf{R}_k, \boldsymbol{\nu}_k \leftarrow \text{LR-ALPCAUS}(\mathbf{Y}_k, \hat{d}_k, I)$
 $\boldsymbol{\Pi}_k^{-1} \leftarrow \text{Diagonal}(\{\nu^{-1} : \nu \in \boldsymbol{\nu}_k\})$ Update current subspace basis estimates
 end for
 $\mathcal{C}_k \leftarrow \{\mathbf{y} \in \mathbf{Y} : \forall j \ \|\mathbf{L}_k \mathbf{L}_k^\dagger \mathbf{y}\|_2 \geq \|\mathbf{L}_j \mathbf{L}_j^\dagger \mathbf{y}\|_2\}$ for $k = 1, \dots, K$ Update current cluster estimates by projection
 end for
 $\mathcal{C}^{(b)} \leftarrow \mathcal{C}_k = \{c_1, \dots, c_N\}$ Collect results from all trials
end for
 $\mathbf{W}_{i,j} \leftarrow \frac{1}{B} |\{\forall b : \mathbf{y}_i, \mathbf{y}_j \text{ are co-clustered in } \mathcal{C}^{(b)}\}|$ for $i, j = 1, \dots, N$ Form affinity matrix of similar clusterings
for $i = 1, \dots, N$ **(in parallel)** **do**
 $\mathbf{Z}_{i,:}^{\text{row}} \leftarrow \mathbf{W}_{i,:}$ with the smallest $N - q$ entries set to zero. Threshold rows of affinity matrix
 $\mathbf{Z}_{:,i}^{\text{col}} \leftarrow \mathbf{W}_{:,i}$ with the smallest $N - q$ entries set to zero. Threshold columns of affinity matrix
end for
 $\mathbf{W} \leftarrow \frac{1}{2} (\mathbf{Z}^{\text{row}} + \mathbf{Z}^{\text{col}})$ Average affinity matrix
 $\mathcal{C} \leftarrow \text{SPECTRALCLUSTERING}(\mathbf{W}, K)$ Final clustering

where $\lambda_i(S_k)$ denotes the i th eigenvalue of S_k assuming $\lambda_1 \geq \dots \geq \lambda_D$. For heteroscedastic data, the eigengap heuristic can break down, making it challenging to determine rank, as shown in Sec. V, Fig. 5. In recent work, [34] develops a parallel analysis algorithm to estimate the rank of a matrix that is consistently shown to work well in the heteroscedastic regime if the data comes from **one** subspace only. It works by creating an i.i.d. Bernoulli ($p = 0.5$) matrix denoted as $\mathbf{M}$ and analyzing the singular values of $\mathbf{M} \odot \mathbf{Y}$ for a matrix of interest $\mathbf{Y}$. One can distinguish what singular values are associated with the signal and noise component of the data through this process. We generalize this line of work and apply it to the union of subspace setting by starting off over-parameterized and adaptively shrinking subspace bases as follows:

$$\forall k \in \{1, \dots, K\} \text{ do} \\ \tilde{\sigma}^{(r)} = \text{SingularValues}(\mathbf{M} \odot \mathbf{Y}_k) \ \forall r \in \{1, \dots, R\} \quad (25)$$

$$\hat{d}_k = \text{smallest } d \text{ that satisfies} \\ \sigma_{d+1}(\mathbf{Y}_k) \leq \alpha\text{-percentile of } \{\tilde{\sigma}_{d+1}^{(1)}, \dots, \tilde{\sigma}_{d+1}^{(R)}\}. \quad (26)$$

This is done for each estimated cluster $\mathbf{Y}_k$ over R random trials where $1 \leq R \ll T$. Here $\sigma_{d+1}(\mathbf{Y}_k)$ denotes the $d + 1$ th singular value of $\mathbf{Y}_k$. We repeat this process for all cluster subsets $\{\mathbf{Y}_1, \dots, \mathbf{Y}_K\}$ after the cluster reassignment update in (17). To reduce computation, we perform this dimension estimate sparingly every 10% of ALPCAUS iterations.

C. Cluster Initialization

For the non-ensemble version of ALPCAUS ($B = 1$), it previously remained to be seen whether there exists an initialization scheme that performs better in expectation than random cluster assignment in the heteroscedastic regime. In recent work, [26] proposes a thresholding inner-product based spectral initialization method (TIPS), designed for homoscedastic data to be used with KSS, where an affinity matrix is generated by

$$\mathbf{W}_{ij} = 1 \text{ if } |\langle \mathbf{y}_i, \mathbf{y}_j \rangle| \geq \tau \text{ and } i \neq j \quad (27)$$

given a thresholding parameter $\tau > 0$ and zero otherwise. Then, the cluster assignments $\mathcal{C} = \{c_1, \dots, c_N\}$ are calculated by applying the spectral clustering method on $\mathbf{W}$. Recall that $\mathbf{y}_i = \mathbf{x}_i + \boldsymbol{\epsilon}_i$. Upon closer analysis, this metric (ignoring absolute value), in expectation gives

$$\begin{aligned} \mathbb{E}[\langle \mathbf{y}_i, \mathbf{y}_j \rangle] &= \mathbb{E}[\langle \mathbf{x}_i, \mathbf{x}_j \rangle] + \mathbb{E}[\langle \mathbf{x}_i, \boldsymbol{\epsilon}_j \rangle] \\ &\quad + \mathbb{E}[\langle \boldsymbol{\epsilon}_i, \mathbf{x}_j \rangle] + \mathbb{E}[\langle \boldsymbol{\epsilon}_i, \boldsymbol{\epsilon}_j \rangle] = \mathbb{E}[\langle \mathbf{x}_i, \mathbf{x}_j \rangle]. \end{aligned} \quad (28)$$

Therefore, the metric is independent of the noise variances, meaning the affinity matrix constructed does not have highly unbalanced, asymmetric edge weights from noisy samples. Thus this metric is more robust to heteroscedastic noise than others such as Euclidean norm where

$$\mathbb{E}[\|\mathbf{y}_i - \mathbf{y}_j\|_2^2] = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2 + D(\boldsymbol{\nu}_i + \boldsymbol{\nu}_j). \quad (29)$$

Clearly, this is inflated by the noise terms $\boldsymbol{\nu}_i$ and $\boldsymbol{\nu}_j$. To note, TIPS initialization is only useful when $B = 1$ for one base

clustering since it is a deterministic initialization; it provably performs better than random initialization as illustrated in Fig. 4. Otherwise, random initialization is used when $B > 1$ to leverage consensus information in the ensemble process.

D. ALPCAUS Convergence

ALPCAUS with a single cluster ($K = 1$) provably converges to local minima since the cost function simplifies to LR-ALPCAUS (14) which has convergence guarantees shown in [17, Thm. 2]. In the multi-cluster setting ($K > 1$), one can guarantee convergence with respect to the cost function in (15) by ensuring two things: 1) that the noise variances are kept above some small threshold value and 2) having a stopping criteria defined as follows. Let the stopping criteria consist of the following cases: 1) if new assignment for $\mathbf{y}_i$ has a new label $c_i = k$ that lowers the cost objective then accept new label, and 2) if $\mathbf{y}_i$ has current label $c_i = k_1$ but the cluster assignment problem (17) has equal cost to some other $c_i = k_2$ then retain the current label k_1 regardless. The argument is a natural extension of the original k-subspaces formulation [25, Thm. 7]. We formally state this result in Thm. 1. We provide an experimental result in the appendix, Fig. 9, to validate this theorem.

Theorem 1. *The ALPCAUS formulation (15) has a sequence of cost function values that converges to some minimum value with cluster assignments being locally optimal assuming a noise variance lower bound parameter $\alpha \in \mathbb{R} > 0$ that lower bounds all ν_i and stopping criteria that rejects repeated assignments of points $\mathbf{y}_i$ and corresponding labels c_i as described earlier.*

Proof. We follow with a proof by contradiction, assume that there is no local convergence of cost function values, i.e., the sequence of cost function values diverges. There are at most N^K ways to partition the N data points into K clusters (finite possible assignments). One of two scenarios arise, either a) the $\mathcal{C}^{(t+1)}$ cluster assignment at the $(t+1)$ iteration will be the same as $\mathcal{C}^{(t)}$, or b) the new assignment $\mathcal{C}^{(t+1)}$ will be different. If scenario a) occurs, then the cost function value will not change meaning the sequence of cost function values has converged. Thus, a contradiction has occurred. The remainder of the proof will focus on scenario b). Let $\mathbf{Y}_k^{(t)}, \mathbf{L}_k^{(t)}, \mathbf{R}_k^{(t)}, \mathbf{\Pi}_k^{(t)}$ be the optimization variables that produce the cost below

$$\text{cost}^{(t)} = \sum_k \frac{1}{2} \|(\mathbf{Y}_k - \mathbf{L}_k \mathbf{R}_k^T) \mathbf{\Pi}_k^{-1/2}\|_F^2 + \frac{D}{2} \log |\mathbf{\Pi}_k|. \quad (30)$$

Note that the cost function value is lower bounded by $DN \log(\alpha)/2$. Because of scenario b), it follows that a new clustering $\mathcal{C}^{(t+1)} \neq \mathcal{C}^{(t)}$ is produced by solving

$$c_i = \arg \min_k \|\mathbf{y}_i - \mathbf{U}_k \mathbf{U}_k^T \mathbf{y}_i\|_2^2 \quad (31)$$

for all data points. Under this setting, there is some $\mathbf{y}_i$ with label $c_i^{(t+1)} \neq c_i^{(t)}$. Due to the stopping criteria, this is only possible if (31) achieved a lower value, meaning the point is actually closer (in an L2 residual sense) to the $c_i^{(t+1)}$ cluster than $c_i^{(t)}$. However, this implies that $\text{cost}^{(t+1)} < \text{cost}^{(t)}$. This is because $\|(\mathbf{Y}_k - \mathbf{L}_k \mathbf{R}_k^T) \mathbf{\Pi}_k^{-1/2}\|_F^2$ in (30) measures the

weighted residual, or sum of squares, for all points that belong in the k th cluster. Recall that a low-rank approximation exists $\mathbf{X}_k \triangleq \mathbf{L}_k \mathbf{R}_k^T$ for $\mathbf{Y}_k$ and $\mathbf{U}_k = \mathbf{L}_k \mathbf{L}_k^\dagger$ is used in (31). However, $\mathbf{X}_k = \mathbf{U}_k \mathbf{U}_k^T \mathbf{Y}_k$ so solving (31) leads to a lower value in (30) since $\|(\mathbf{Y}_k - \mathbf{L}_k \mathbf{R}_k^T) \mathbf{\Pi}_k^{-1/2}\|_F^2$ measures the same thing (ignoring the weight). Therefore, a contradiction is achieved if scenario b) occurs. From scenario a) and b), it follows that the cost function value (30) does not diverge. Instead, due to finite possibilities of the N points being assigned to the K total clusters, stopping criteria that rejects repeated assignments, a non-increasing cost function that is bounded below, it follows that the ALPCAUS cost function (30) has a sequence of cost function values that will converge to a local minima with cluster assignments being locally optimal. $\square$

V. EXPERIMENTS

A. Synthetic Experiments

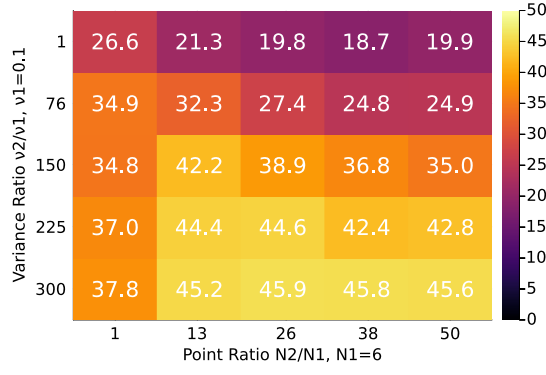
1) *Experimental Setup:* We generated a synthetic dataset consisting of $K = 2$ clusters each of dimension $d = 3$ derived from random subspaces in $D = 100$ dimensional ambient space. Each cluster consisted of two data groups with group 1 containing $N_1 = 6$ samples, to explore the data constrained regime, with noise variance $\nu_1 = 0.1$ per cluster. For group 2, we varied the N_2 samples and noise variance ν_2 . We performed cross validation for hyperparameters in any algorithms that require it with a separate training set. We used the estimated clusters from each clustering algorithm and the known clusters to calculate clustering error using the Hungarian algorithm [35] to deal with the problem of label permutations. More concretely, clustering error is defined as

$$\text{clustering error} = \frac{100}{N} (1 - \max_{\pi} \sum_{i,j} \mathbf{Q}_{\pi(ij)}^{\text{out}} \mathbf{Q}_{ij}^{\text{true}}) \quad (32)$$

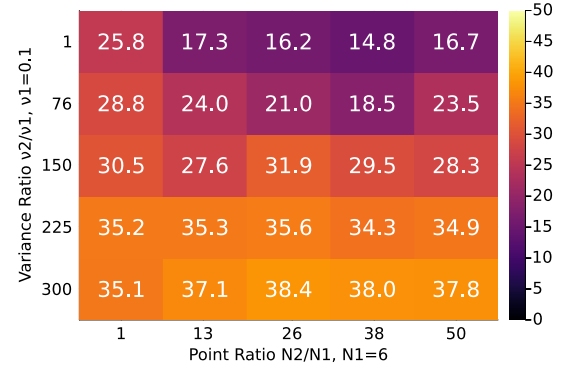
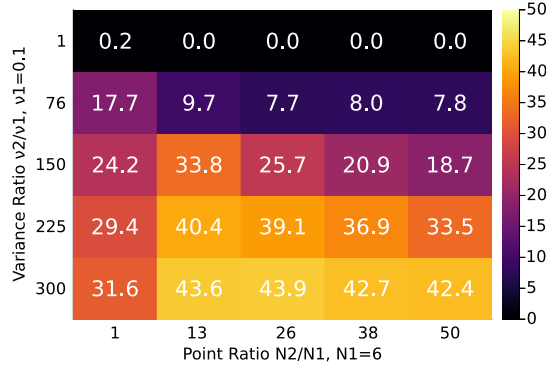
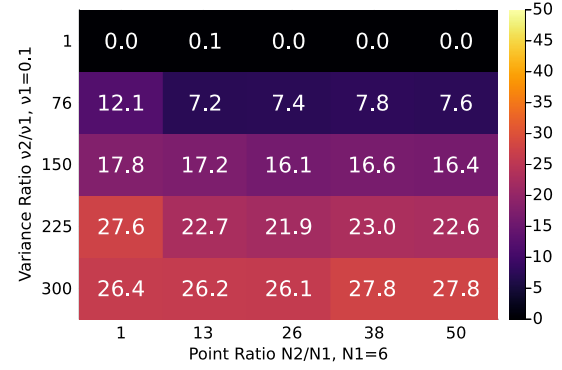
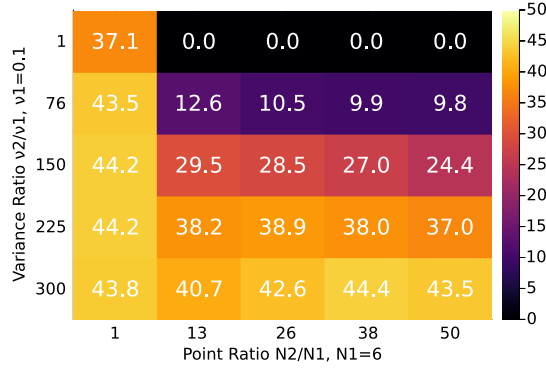
where π is a permutation of cluster labels, and $\mathbf{Q}^{\text{out}}$ and $\mathbf{Q}^{\text{true}}$ are output labels and ground truth labelings of the data where the (i, j) th entry is one if point j belongs to cluster i and zero otherwise.

We tried different values of noise variances and data samples and computed average clustering error out of 100 trials. Each trial had different noise, basis coefficients, and subspace basis realizations for the clusters. We compared various subspace clustering algorithms such as K -Subspaces (KSS) [14], Ensemble K -Subspaces (EKSS) [16], Subspace Clustering via Thresholding (TSC) [36], and Doubly Stochastic Sparse Subspace Clustering (ADSSC) [37]. Applying general non-subspace clustering methods such as K -means [38] resulted in poor clustering error so results of these methods are generally not shown.

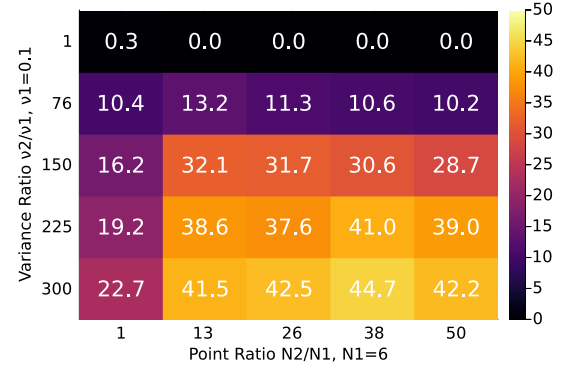
2) *Clustering Error:* Table 2h shows the clustering quality of these algorithms for synthetic heteroscedastic data. Fig. 2 complements Table 2h by exploring the complete heteroscedastic landscape with a visual representation that is easier to understand. We define a method called “noisy oracle” where an oracle uses the true cluster assignments to apply PCA on the low noise data only per each cluster. Using these subspace basis estimates of each cluster, the oracle performs cluster assignments using (17). This oracle reference provides a kind



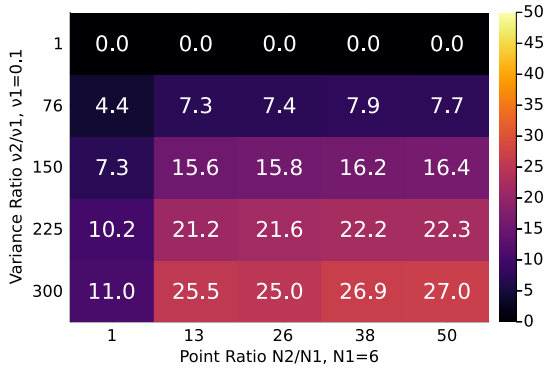
(a) KSS (TIPS) mean clustering error.

(b) ALPCAUS ($B = 1$, TIPS) mean clustering error.(c) EKSS ($B=128$) mean clustering error.(d) ALPCAUS ($B=128$) mean clustering error.

(e) TSC mean clustering error.



(f) ADSSC mean clustering error.



(g) Noisy oracle mean clustering error.

ν_2/ν_1	1	1	300	300	150	225	76
N_2/N_1	1	50	1	50	26	13	38
Reference Level							
Noisy Oracle	0.0	0.0	11.0	27.0	15.8	21.2	7.9
Method Comparisons							
KSS	26.6	19.9	37.8	45.6	38.9	44.4	24.8
EKSS ($B = 128$)	0.2	0.0	31.6	42.4	25.7	40.4	8.0
ADSSC	0.3	0.0	22.7	42.2	31.7	38.6	10.6
TSC	37.1	0.0	43.8	43.5	28.5	38.2	9.9
ALPCAUS ($B = 1$)	25.8	16.7	35.1	37.8	31.9	35.3	18.5
ALPCAUS ($B = 128$)	0.0	0.0	26.4	27.8	16.1	22.7	7.8

(h) Clustering error (%) results for heteroscedastic landscape. Visual results included in Fig. 2. Columns consist of the 4 corners and the 3 diagonal elements of the heatmaps in Fig. 2.

Fig. 2: Clustering error over the heteroscedastic landscape for various subspace clustering algorithms.

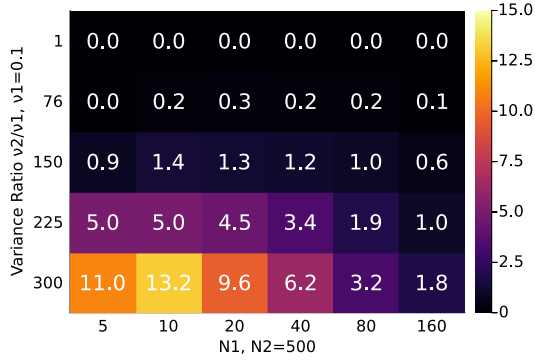


Fig. 3: Percentage difference (%) of ALPCAUS clustering error subtracted from EKSS while good data amount varies.

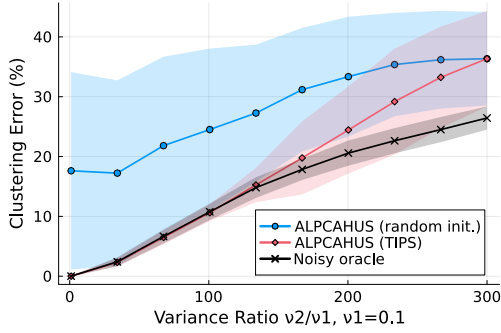


Fig. 4: Clustering error (%) for TIPS initialization scheme vs. random initialization for the ALPCAUS method ($B = 1$).

of lower bound on realistic clustering performance since it can approximate subspace bases separately given the true labeling.

ALPCAUS ($B = 1$) with TIPS initialization achieved lower clustering error than KSS with TIPS initialization. For the ensemble methods ($B > 1$), both EKSS and ALPCAUS significantly improved. However, EKSS still had higher clustering error in more heteroscedastic regions. Meanwhile, ALPCAUS remained very close to the noisy oracle, indicating it more accurately estimated the underlying true labeling. Compared to other methods like TSC and ADSSC, ALPCAUS generally outperformed them with one exception for ADSSC ($N_2 = 1, v_2/v_1 = 300$). Here, ADSSC performed better than ALPCAUS when the number of noisy points is roughly equal to the number of good points. Overall, the ensemble version of ALPCAUS was generally more robust against heteroscedasticity compared to other subspace clustering algorithms. To summarize, we explored the effects of data quality and data quantity on the heteroscedastic subspace clustering estimates in different situations. To our knowledge, this is the first systematic analysis of heteroscedasticity effects on subspace clustering quality in the literature.

3) *Effects of Good Data*: Additionally, we explored the effects of good data quantity on the subspace clustering problem to understand at what point it becomes advantageous to use ALPCAUS over KSS methods. We fixed $v_1 = 0.1, N_2 = 500$ and vary N_1, v_2 . We report the percentage difference of EKSS clustering error (%) - ALPCAUS clustering error (%), i.e., $\text{error}_{\text{EKSS}} - \text{error}_{\text{ALPCAUS}}$, meaning higher values indicate our

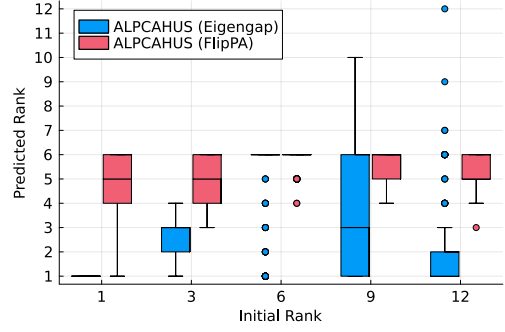


Fig. 5: Adaptive rank estimation using eigengap heuristic and proposed FlipPA approach (true rank $d = 6$).

method is better. Figure 3 shows that ALPCAUS performed better than EKSS even up to $N_1 = 160$ which represents about 33% of the total data. ALPCAUS never performed worse than EKSS but the advantage gap narrowed as N_1 increased. Thus, there is a wide range of conditions for which ALPCAUS is preferable to homoscedastic methods.

4) *Clustering Initialization*: Section IV proposed using the TIPS initialization scheme in this heteroscedastic context since the dot product metric used in constructing the affinity matrix in (28) is provably robust to heteroscedastic noise. We fixed points groups N_1, N_2 and varied v_2 while keeping $v_1 = 0.1$. We included the noisy oracle to establish a realistic performance baseline. Figure 4 shows that TIPS initialization ALPCAUS ($B = 1$) outperformed random initialization for the non-ensemble ALPCAUS method ($B = 1$) across the entire heteroscedastic landscape. Thus, TIPS should always be used for the non-ensemble methods for clustering quality. We did not include the ensemble version of ALPCAUS ($B > 1$) because the ensemble process inherently takes advantage of random initialization to achieve a different labeling each trial.

5) *Rank Estimation of Clusters*: In Sec. IV, we proposed to use random sign flipping to adaptively find and shrink the subspace dimension of clusters when subspace dimension is unknown. For Fig. 5 we synthetically generated subspaces with true rank $d = 6$ and explored how the initial rank parameter affects the ability to estimate the true rank over 100 trials. We compared this approach against the eigengap heuristic by using ALPCAUS with both adaptive rank schemes and report the estimated rank values from the final clustering. The eigengap approach consistently underestimated the rank regardless of the initial value, except when given the true rank value $d = 6$. Our sign flipping approach provided much better rank estimates with much smaller variances between trials.

B. Real Data Experiments

1) *Quasar Flux Data*: We investigated quasar spectra data from the Sloan Digital Sky Survey (SDSS) Data Release 16 [39] using its DR16Q quasar catalog [40]. Each quasar has a vector of flux measurements across wavelengths that describes the intensity of observing that particular wavelength. In this dataset, the noise is heteroscedastic across the sample space (quasars) and feature space (wavelength), but we focused on a subset of

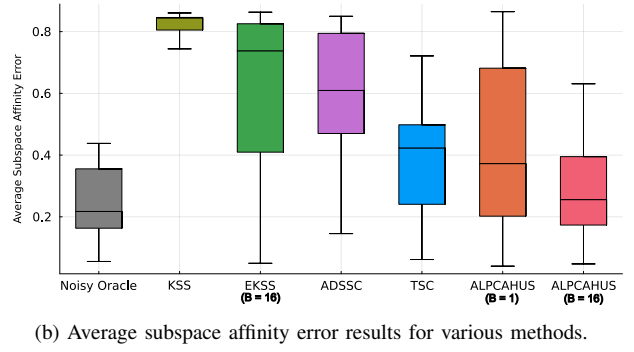
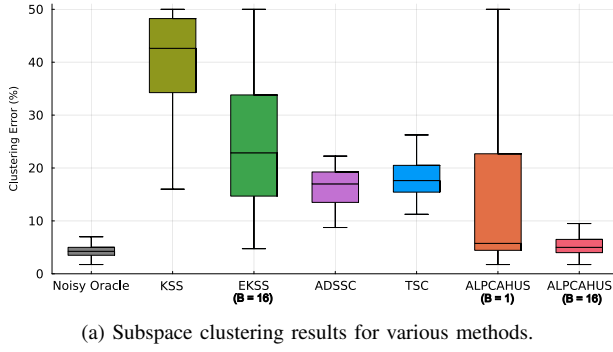


Fig. 6: Experimental results of quasar flux data for subspace clustering and learning. Methods involving a single run, i.e., KSS and ALPCAUS ($B = 1$), use the TIPS initialization scheme.

Methods	Time (ms)	Memory (MiB)	Mean Subspace Error	Mean Clustering Error (%)
KSS	149.3	53.0	0.80	38.5
EKSS (B=16)	181.4	212.6	0.62	23.8
ADSSC	125.7	17.5	0.62	21.1
TSC	32.3	16.5	0.40	18.2
ALPCAUS (B=1)	148.5	126.7	0.43	14.4
ALPCAUS (B=16)	210.7	678.2	0.28	6.3

TABLE I: Subspace clustering results on quasar flux data. The KSS and ALPCAUS ($B = 1$) methods use TIPS initialization.

data that is homoscedastic across wavelengths and heteroscedastic across quasars. The noise for each quasar is known given the measurement devices used for data collection [39]. In Fig. 7, a subset of the spectra data is shown for illustrative purposes to compare the visual differences in data quality. We preprocessed the data (filtering, interpolation, centering, and normalization) based on the supplementary material of [41]. We form clusters in these experiments by considering quasars with different properties, namely, two quasar groups ($K = 2$) with different redshift values ($z_1 = 1.0 - 1.1$, $z_2 = 2.0 - 2.1$). Additionally, for the second group, we queried for broad absorption lines (BAL) type quasars. Since we downloaded the data using separate queries, we know which points belong to which cluster group, meaning we can compute clustering error for comparison purposes. We isolated a training set (5000 samples total) to learn any model parameters and used the rest (5000 samples total) for a test dataset. We formed trials by randomly selecting 400 quasar spectra samples per group and performed 50 total trials to report clustering error. For rank estimation, we used the noisy oracle approach and found that there is no improvement to clustering quality for rank values greater than $\hat{d} = 3$, so we used this value for any applicable algorithms.

Figure 6a shows clustering error for this quasar spectra multi-cluster data. The ensemble version of ALPCAUS was very close in clustering quality to the noisy oracle, suggesting that it accurately learned the subspace bases while clustering the data groups. The non-ensemble version of ALPCAUS ($B = 1$) had large variances in clustering error, indicating some challenges in getting close to the optimal cost function minima with this data. However, based on median values, it

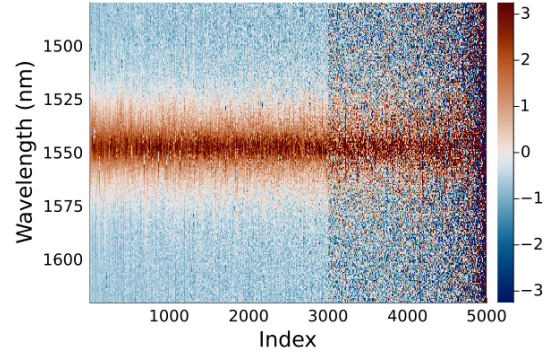


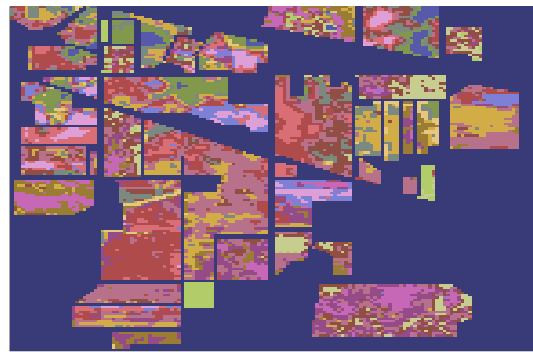
Fig. 7: Sample data matrix of quasar flux measurements across wavelengths for each (column-wise) sample.

still managed to get a lower error than KSS, ADSSC, and TSC. Additionally, Fig. 6b shows the average subspace affinity error of these methods after applying LR-ALPCAUS to the clusterings for all methods. ALPCAUS better learned the subspace bases than the other methods when $B = 16$, which was chosen as the smallest value that had small run-to-run variances while not taking significantly longer to compute. Both of these results show the benefit of developing heteroscedastic specific algorithms in the subspace clustering context. Table I reports median time complexity and memory requirements along with mean clustering error and mean subspace error. In this data instance, the ensemble method gets close in time to the non-ensemble methods due to our multi-threaded implementation. Yet, because of the multi-threaded implementation, the memory requirements are larger for the ensemble method as opposed to the non-ensemble method. Overall, relative to other clustering methods, ALPCAUS is competitive time-wise at the cost of increased memory requirements.

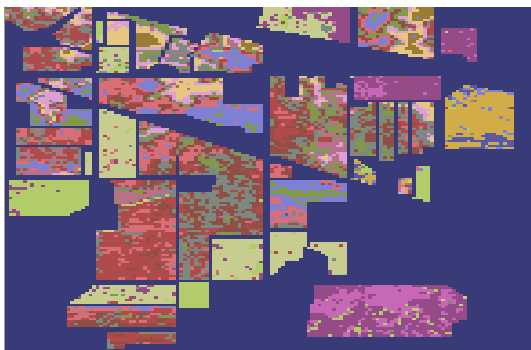
2) *Indian Pines Data*: Additionally, we investigated hyperspectral image (HSI) segmentation data called *Indian Pines* [42]. This image of size 145×145 contains $D = 200$ reflectance bands for each pixel that is around the $0.4 - 2.5 \mu\text{m}$ wavelength. HSI data is known to be very noisy due to thermal effects, atmospheric effects, and camera electronics, leading to works that study per-pixel noise estimation in hyperspectral imaging [43]. In total, there are $K = 16$ classes composed of alfalfa, corn, grass, wheat, soybean, and others. There is one



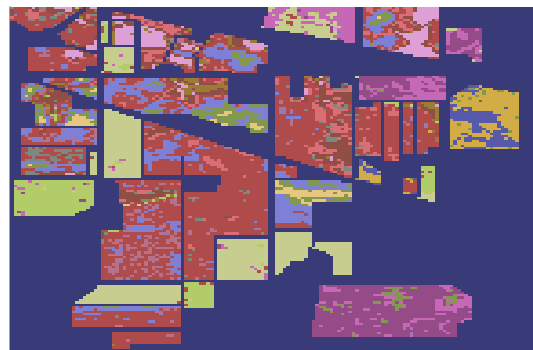
(a) Ground truth labels for *Indian Pines* dataset. NC means no class available (background).



(b) K -means results with clustering error = 77% and mIOU = 16%.



(c) EKSS ($B = 32, T = 3, q = K$) results with clustering error = 64% and mIOU = 27%.



(d) ALPCAUS ($B = 32, T = 3, q = K$) results with clustering error = 53% and mIOU = 31%.

Fig. 8: Experimental results of *Indian Pines* data for subspace clustering.

additional class (background) that is withheld from clustering as commonly done [33]. Therefore, $N = 10,249$ instead of $N = 145 \times 145$. In other works, researchers have found that $\hat{d} = 5$ is an appropriate rank parameter since it covers 95% of the cumulative variance [33]. A subset of the data matrix $Y \in \mathbb{R}^{200 \times 10249}$ is split into 2,500 samples to learn model parameters. From this, the learned subspaces are applied on Y to cluster all data so that the heatmaps can be compared against the ground truth. The ground truth image is found in Fig. 8a. For perspective, randomly guessing would lead to 94% clustering error due to $K = 16$ clusters. For consistency, the Hungarian algorithm is again applied to the clustering results to make it easier to compare against the ground truth image.

We compare against K -means in Fig. 8b to illustrate the difficulty of the problem. ALPCAUS results are shown in Fig. 8d next to EKSS results in Fig. 8c. Clustering error is reported for these algorithms along with mean IOU to measure the similarity between the images. ALPCAUS achieved the lowest clustering error and highest mIOU (53% / 31%) relative to the other approaches such as K -means (77% / 16%) and EKSS (64% / 27%). We note that our ALPCAUS results are similar to [33] even though, in that work, they instead treat the noise as homoscedastic and instead model the signal as heteroscedastic. Further, the EKSS results on this dataset are similar to the homoscedastic PCA approach used in [33]. This indicates some utility in modeling heteroscedasticity in

hyperspectral images. Yet, the reflectance bands themselves also appear to be heteroscedastic with some bands being noisier than others [44]. Thus, developing a method that is doubly heteroscedastic with respect to both the samples and features is an interesting direction of future work.

For reference, the SOTA result is about 10% misclassification rate in a *classification* setting (i.e., not clustering) [45]. In a clustering setting, recent works such as [46] have achieved a 40% clustering error which is 13% lower than our results. We do not claim SOTA results with HSI data, only that heteroscedastic union of subspace modeling improved results over homoscedastic union of subspace modeling in finding reflectance band groups with similar characteristics.

VI. CONCLUSION

This paper proposed ALPCAUS, a subspace clustering algorithm that can find subspace data clusters whose points contain heteroscedastic noise. For future work, a union of manifolds generalization could possibly be useful. For example, Alzheimer's disease patients often have different resting state functional MRI activations than cognitively normal individuals [47], so one could consider these different classes to belong to different manifolds in the ambient space. Previous research has shown manifold learning to more correctly model temporal dynamics than subspace modeling in this domain [48]. Another direction of future work could be to consider heteroscedasticity

across the feature space. For example, one might be interested in biological sequencing of different species with similar genes where the gene marker counts naturally follow heteroscedastic distributions [49]. These generalizations are nontrivial so it is left for future work. While our implementation of ALPCAHUS benefited from parallelization, it needed more computation time and memory than TSC. Because of this, there is an opportunity to further optimize our code base to remove certain matrix multiplication operations and instead use single vector dot products to reduce memory. Furthermore, since our subspace basis step requires an SVD computation for an initial low-rank estimate, we could use the Krylov-based Lanczos algorithm [50] to further reduce time and memory for each trial leading to more significant improvements in memory usage and time. Overall, ALPCAHUS was more robust towards heteroscedastic noise than other clustering methods, making it easier to identify correct clusterings under this kind of noise model.

VII. ACKNOWLEDGMENTS

This work was supported in part by NSF CAREER Grant CCF-1845076 and NSF Grant CCF-2331590. There are no financial conflicts of interest in this work. We thank David Hong for suggesting heteroscedastic data applications such as astronomy spectra used in this work.

VIII. APPENDIX

A. ALPCAHUS Convergence Experiment

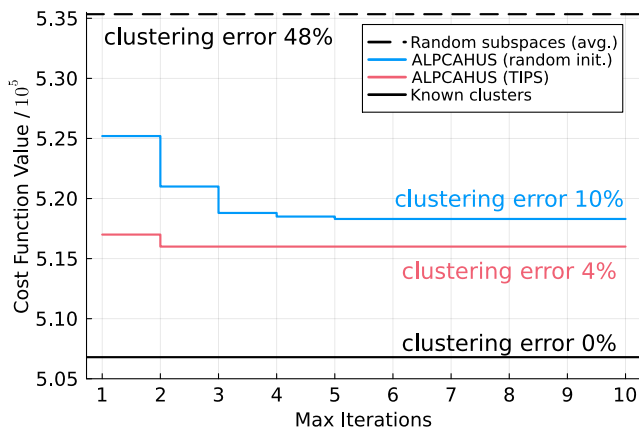


Fig. 9: ALPCAHUS ($B = 1$) cost function value convergence plot following Thm. 1. Astronomy spectra data from Fig. 6a used. For convenience, upper and lower bounds of cost function values are provided by using random subspaces and estimated LR-ALPCAH subspaces (oracle approach) respectively.

This section focuses on providing empirical verification of our ALPCAHUS convergence theorem in Thm. 1. We implement the noise variance lower bound parameter α to bound noise variance estimation and stopping criteria that rejects repeated assignments of points. The quasar spectra flux data from SDSS (DR16Q catalog) is used in this experiment. For simplicity, only one base trial ($B = 1$) is run to include results from the randomly initialized ensemble ALPCAHUS method (generally when $B > 1$) and the TIPS initialized

ALPCAHUS (generally when $B = 1$). In Fig. 9, the cost function value over iterations is plotted for the ALPCAHUS method. Upper and lower bound cost function values are provided by 1) using random initialized subspaces as the “true” subspaces and 2) estimated subspaces by LR-ALPCAH (oracle approach) respectively. As can be observed, ALPCAHUS converges relatively quickly with only a few iterations. We note that in this implementation the stopping criteria does not terminate the algorithm, only rejects the assignments from changing. Hence, ALPCAHUS is run for a fixed 10 iterations even though convergence is achieved earlier for both versions.

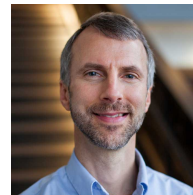
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Javier Salazar Cavazos (Student Member, IEEE) received dual B.S. degrees in electrical engineering and mathematics from the University of Texas at Arlington, Arlington, TX, USA, 2020 and the M.S. degree in electrical engineering from the University of Michigan, Ann Arbor, MI, USA, 2023. He is currently working toward the Ph.D. degree in electrical and computer engineering working with Professor Laura Balzano and Jeffrey Fessler both with the University of Michigan, Ann Arbor, MI, USA. His main research interests include signal and image processing, machine learning, deep learning, and optimization. Specifically, current work focuses on topics such as low-rank modeling, clustering, and Alzheimer's disease diagnosis and prediction from MRI data.



Jeffrey A. Fessler (Fellow, IEEE) received the B.S.E.E. degree from Purdue University, West Lafayette, IN, USA, in 1985, the M.S.E.E. degree from Stanford University, Stanford, CA, USA, in 1986, and the M.S. degree in statistics and the Ph.D. degree in electrical engineering from Stanford University, in 1989 and 1990, respectively. From 1985 to 1988, he was a National Science Foundation Graduate Fellow with Stanford. He is currently the William L. Root Professor of EECS with the University of Michigan, Ann Arbor, MI, USA. His research focuses on various aspects of imaging problems, and he has supervised doctoral research in PET, SPECT, X-ray CT, MRI, and optical imaging problems. In 2006, he became a Fellow of the IEEE for contributions to the theory and practice of image reconstruction.



Laura Balzano (Senior Member, IEEE) received the Ph.D. degree in ECE from the University of Wisconsin, Madison, WI, USA. She is currently an Associate Professor of electrical engineering and computer science with the University of Michigan, Ann Arbor, MI, USA. Her main research focuses on modeling and optimization with big, messy data – highly incomplete or corrupted data, uncalibrated data, and heterogeneous data – and its applications in a wide range of scientific problems. She is an Associate Editor for the IEEE Open Journal of Signal Processing and SIAM Journal of the Mathematics of Data Science. She was a recipient of the NSF Career Award, the ARO Young Investigator Award, the AFOSR Young Investigator Award. She was also a recipient of the Sarah Goddard Power Award at the University of Michigan.

Paper Title:

ALPCAHUS: Subspace Clustering for Sample-wise Heteroscedastic Data

Authors:

Javier Salazar Cavazos[✉], *Graduate Student Member, IEEE*,

Jeffrey A. Fessler[✉], *Fellow, IEEE*, and Laura Balzano[✉], *Senior Member, IEEE*

Dear IEEE TSP Editor in Chief,

We would like to submit a new submission titled “ALPCAHUS: Subspace Clustering for Sample-wise Heteroscedastic Data” where we extend ideas in a previous TSP submission titled “ALPCAH: Subspace Learning for Sample-wise Heteroscedastic Data” (<https://doi.org/10.1109/TSP.2025.3537867>). Previously, we worked on learning a subspace basis when data follows a heteroscedastic distribution. In this work, we generalize and extend this to the union of subspace setting to do general subspace clustering. In the first original submission, TSP.2025.3537867 included both subspace learning and subspace clustering (ALPCAH and ALPCAHUS). Once reviews came in, we took the reviewers’ advice and split the topics in the paper due to page limitations. It was difficult to keep both topics in the same work and be able to fully address reviewer comments. Because of this, the revised manuscript in TSP.2025.3537867 focused on subspace learning only with the ALPCAH method.

In this submission, we focus on the subspace clustering method ALPCAHUS and develop it further after the preliminary feedback in TSP.2025.3537867. We would like to ask to work with the same associate editor, Dr. Renato Cavalcante (renato.cavalcante@ieee.org), since part of this submission includes material seen in the first submission of TSP.2025.3537867. Moreover, even though it may be challenging to get all of the same reviewers from TSP.2025.3537867, we feel it would be beneficial to get as close as possible to the same reviewers since they are familiar with our work due to their initial reading of TSP.2025.3537867. Thank you for your considerations in this request given that this is not a totally “new” submission.

Regards,

ALPCAHUS Authors

IX. RELEVANT RESPONSES FROM TSP.2025.3537867:

One similar theme noted by most reviewers was that our work included too much material in both subspace learning and subspace clustering, making it challenging to be complete within the page limit. The previous version was already at the 16 page absolute limit, posing challenges in incorporating reviewer comments. We reached out to the associate editor after reviewer 2 suggested that it would be preferable to separate subspace learning and subspace clustering to give each topic more attention and more detailed discussion. By doing this, we are able to better address reviewer comments and add extra details that make our work more clear. **The comments below relate to the subspace clustering aspect of TSP.2025.3537867.**

R1.C2 - LR-ALPCAUS is presented too briefly. The authors should explain where the steps of the algorithm come from. Same thing for the ALPCAUS algorithm.

Derivations have been expanded upon to be more clear where the steps of the algorithm come from. For example, (19), (19),(20) give greater detail on the alternating minimization updates. Alg. ALPCAUS now contains code comments that make the steps in the algorithm more clear.

R1.C10 - In eq (21), $C, \Pi, \mathcal{L}, \mu$ should be defined more explicitly.

The subspace clustering part has been removed as discussed in the overall reply to reviewers; we will take this feedback to improve our ALPCAUS submission.

R1.C12 - In Algorithm ALPCAUS, can you detail what you mean by "SPECTRALCLUSTERING"? (this spectral clustering algorithm is not detailed in the paper)

Spectral Clustering entails constructing a $N \times N$ affinity matrix for N data points. The graph Laplacian is computed and the rows of the eigenvectors of the Laplacian contain clustering structure. K-means is used on the row embeddings of the eigenvectors of the graph Laplacian. The problem is essentially finding the "cut" in the graph that has minimum cost to separate out clusters from the affinity matrix. For more details, see this tutorial paper (<https://doi.org/10.48550/arXiv.0711.0189>). This comment is no longer applicable here now that ALPCAUS will be submitted later in a separate paper, but we will be sure to expand on spectral clustering for that submission.

We have expended on spectral clustering in Sec. III-A1.

R1.C14 - In Fig 11, can you add simple baselines outside from the subspace-based methods? e.g. k-means, spectral clustering, etc. so that we can see the difficulty of the problem.

From preliminary experiments, k-means performed very poorly with 0.45 clustering error, so not much better than random guessing which is why we did not include it originally. This comment no longer applies for this submission, since ALPCAUS was removed, but we appreciate the feedback and will include more baselines in our ALPCAUS submission.

R2.C1 - While the paper reaches the number of pages limitation, I believe that many crucial information is missing from the manuscript to meet the standards of top quality research. In my opinion, adding sufficient information while respecting the final length limitation is impossible. The author should either split this paper in two with the subspace learning in one and the subspace clustering in the other (the contributions are probably enough). Or they could also submit their work in a journal without such length limitation . The only other solution I can see would be to drastically reduce the numerical experiment section, but this would significantly decrease the quality of the paper in my opinion (even though this section needs some significant improvements).

We agree. To give each section enough attention and fill in crucial information, we conferred with the AE and decided to focus on subspace learning for this revised submission. The AE agreed that this submission can be shifted towards subspace learning (ALPCA) and the subspace clustering can be submitted later (ALPCAUS). This change allows us to remain within the page limit and expand on certain sections.

R2.C2 - As is, the paper feels quite unfinished. Not all notations are introduced. Many things are not properly explained, i.e., not all is given to simply understand and reproduce the computations without being an absolute expert on every details of the present manuscript. In particular, I feel that the derivations of the algorithms should be properly provided, not only the resulting algorithms. Some statements do not really feel that they are properly justified. For instance, "ALPCAUS [...] overcomes these limitations". I do not feel that the authors actually provide guarantees for ALPCAUS to overcome these limitations. Furthermore, from my understanding, the strategies provided to overcome these limitations are not specific to the proposed model and could be employed in the other cases as well.

Other reviewers agreed that the notation needed more work and they provided detailed comments on specific areas that needed improvement. We have taken this advice to refine notation throughout the paper.

R2.C4 - Notice that there also exists : Collas, A., et al (2021). Probabilistic PCA from heteroscedastic signals: geometric framework and application to clustering. IEEE TSP. It deals with a similar model of heteroscedastic data. Even though it is a bit different, it might be of interest in such a review section. It also seems particularly interesting since it also considers clustering.

We added a discussion about other heteroscedastic models including that paper to Sec. III-C.

R2.C5 - This section is a bit better in my opinion. However, there are still the same type of issues. In particular, " Y_k represent a submatrix whose columns consist of points near the k subspace": what does this mean ?? I really do not understand "points near the k subspace". Is it points belonging to the k subspace ? I definitely need a proper formal definition. There's some typos around (9) with k vs $k(i)$. I also believe that a formal definition of the cluster assignments C is needed to improve clarity.

The notation Y_k meant points belonging to the subspace k that were collected into a matrix denoted as Y_k . We appreciate the feedback regarding notation related to ALPCAUS; we will work on a proper formal definition for cluster assignments C . ALPCAUS was removed from this submission so these notational comments are no longer applicable to this revision.

R2.C6 - The description of spectral clustering methods should be completely reworked. As is, it is quite hard to follow. The idea of how it works is not introduced at the start so one has to follow the technical description while fighting to understand what is going on. Variables such as the "coefficient matrix" are not really introduced so one kind of has to guess their use. The constrained set S_P is written the same way as the SCM in the previous section, which is quite confusing. Moreover, to make this less abstract, some actual examples of such sets would be appreciated. It would also be nice to have a proper definition of the spectral embedding of the affinity matrix. The description of iterative methods, which, I believe, corresponds more to what is proposed in the present paper, feels a bit short to provide some kind of a review of the state of the art.

Thanks for the feedback regarding spectral clustering and other clustering methods used in the original submission; others have mentioned the same. We will use these comments to improve the later subspace clustering submission and provide more detail on spectral clustering and others. We have also refined the notation so that S corresponds to matrices whereas $\mathcal{S}$ corresponds to sets.

R2.C10 - Here, I basically have the same concerns as before, i.e., there is a lack of details to be able to actually reproduce the maths leading to the method, some more details about the initialization choices would be appreciated, etc.

Subspace clustering section has been removed from this submission. We appreciate the feedback regarding ALPCAUS and will use it to improve the later submission.

R2.C13 - Concerning the clustering, I really do not understand the clustering error of the so-called noisy oracle. The first problem is that the clustering error is actually not formally defined. It is said that it corresponds to the number of incorrectly grouped point. On the other hand, it is said that the noisy oracle refers to the case where one knows the cluster assignments. From there, how is it possible for the noisy oracle to have a clustering error different from 0 ? Probably, a more formal description of the noisy oracle would also help. One other thing that I wonder about this so-called baseline is that the authors say that they only use the good quality data. So they only use 6 samples, which are vectors with 100 entries. Thus, there is 600 values to learn the parameters. The dimension of the corresponding Grassmann manifold is 291. There is thus 291 parameters to estimate for the subspace. 600 values to estimate them feels quite low.

We agree with this comment; we will formalize clustering error and noisy oracle in the future ALPCAUS submission.

R3.C2 - Cluster Initialization. The authors mention that TIPS initialization and random initialization are used in different cases, but there is no discussion on how sensitive the results are to the initialization scheme and the hyperparameters used throughout. Given that initialization can have a significant impact on clustering algorithms, a more detailed discussion and perhaps a robustness study on initialization methods would be valuable, so that it will provide some more guidance and instructions on potential real applications in the future.

We agree that more discussion should be given to initialization which is now possible since we are splitting subspace learning and subspace clustering. The subspace clustering aspect of the paper has been removed so that we can better address reviewer comments while being within the page limit.